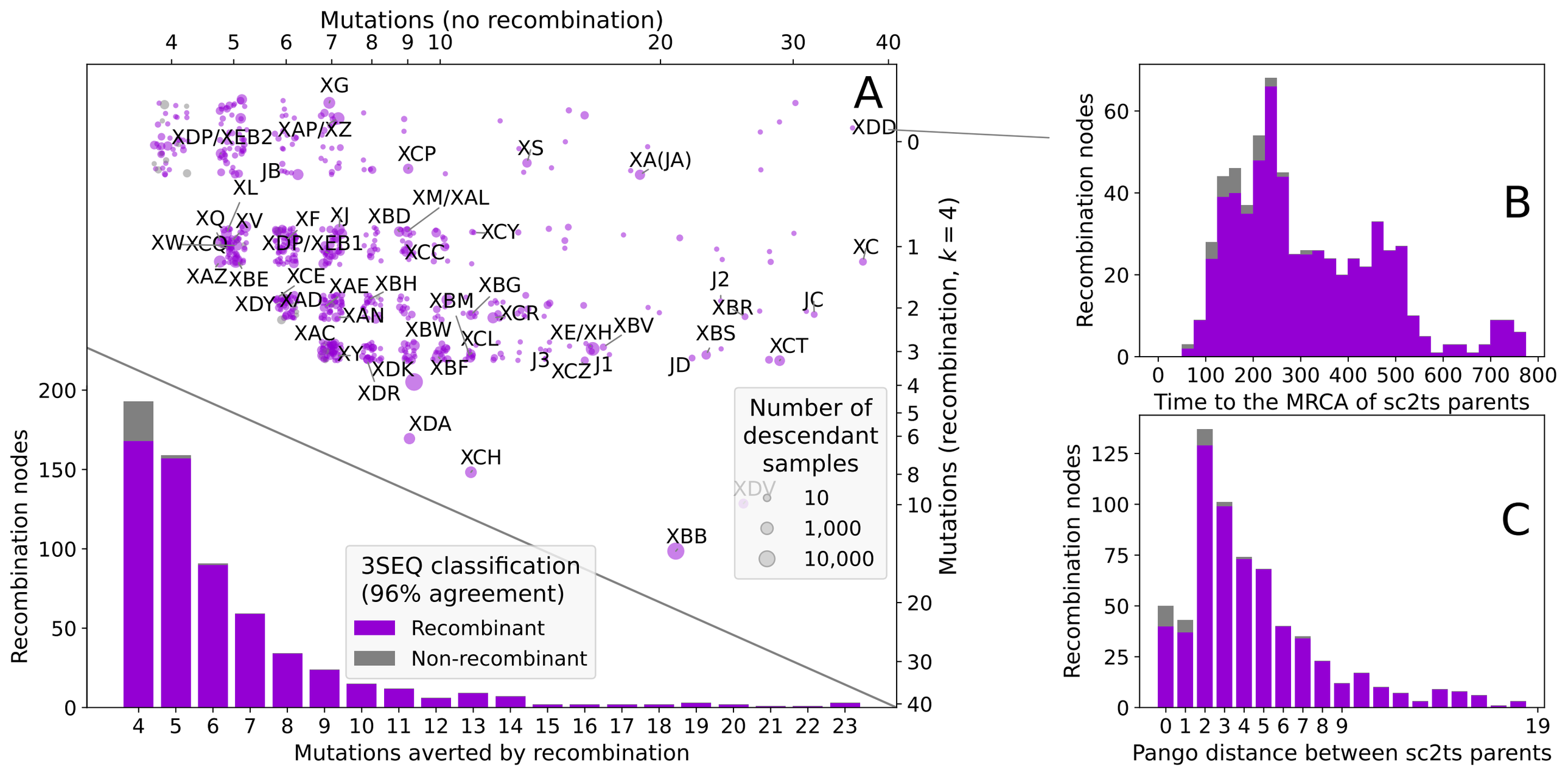

A
Mutations (no recombination)
4 5 6 7 8 9 10 20 30 40
Mutations (recombination, k = 4)
0 1 2 3 4 5 6 8 10 20 30 40
Recombination nodes
0 50 100 150 200
Mutations averted by recombination
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
XG
XDP/XEB2
XAP/XZ
XCP
XS
XA(JA)
XDD
JB
XL
XM/XAL
XQ
XV
XF
XJ
XBD
XCY
XW
XCQ
XDP/XEB1
XCC
XC
XAZ
XBE
XCE
XAE
XBH
XBG
J2
XDY
XAD
XBM
XBR
JC
XAN
XCR
XAC
XBW
XCL
XE/XH
XBV
XBS
XCT
XY
XDK
XBF
J3
XCZ
J1
JD
XDR
XDA
XCH
XBB
XDV
3SEQ classification
(96% agreement)
Recombinant
Non-recombinant
Number of
descendant
samples
10
1,000
10,000
B
Recombination nodes
0 20 40 60
Time to the MRCA of sc2ts parents
0 100 200 300 400 500 600 700 800
C
Recombination nodes
0 25 50 75 100 125
Pango distance between sc2ts parents
0 1 2 3 4 5 6 7 8 9 19